

Sensitive troponins

- what are they, what does it mean and what are the clinical implications

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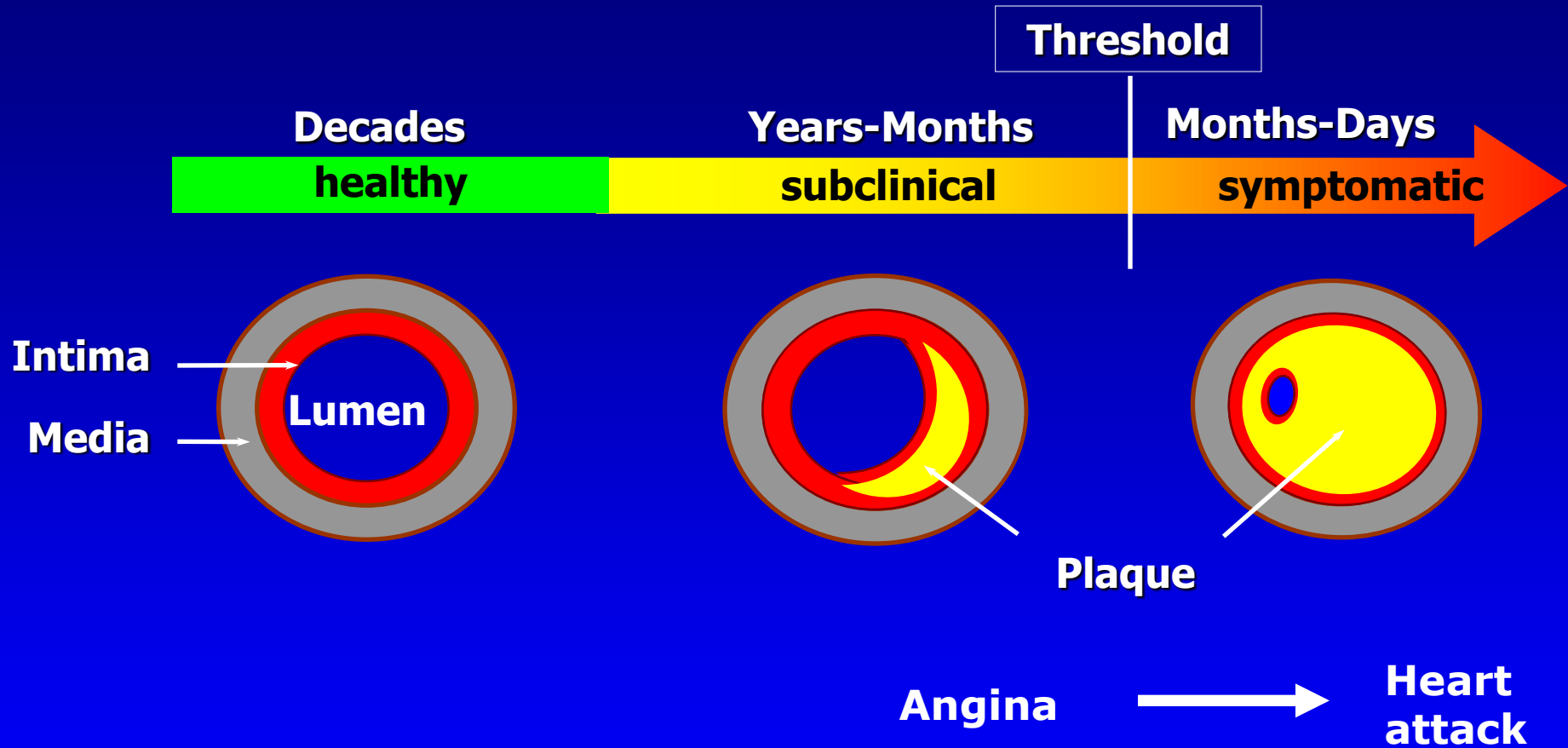
Conflicts of interest

- Member NICE Diagnostics Advisory Committee and Evidence Resources Reference Panel
- National Clinical Lead National Laboratory Medicine Catalogue
- Over the past 25 years advisor/speaker/chair for every manufacturer of cardiac biomarkers that has existed/does exist. Currently advisory Board member for Siemens and Philips.

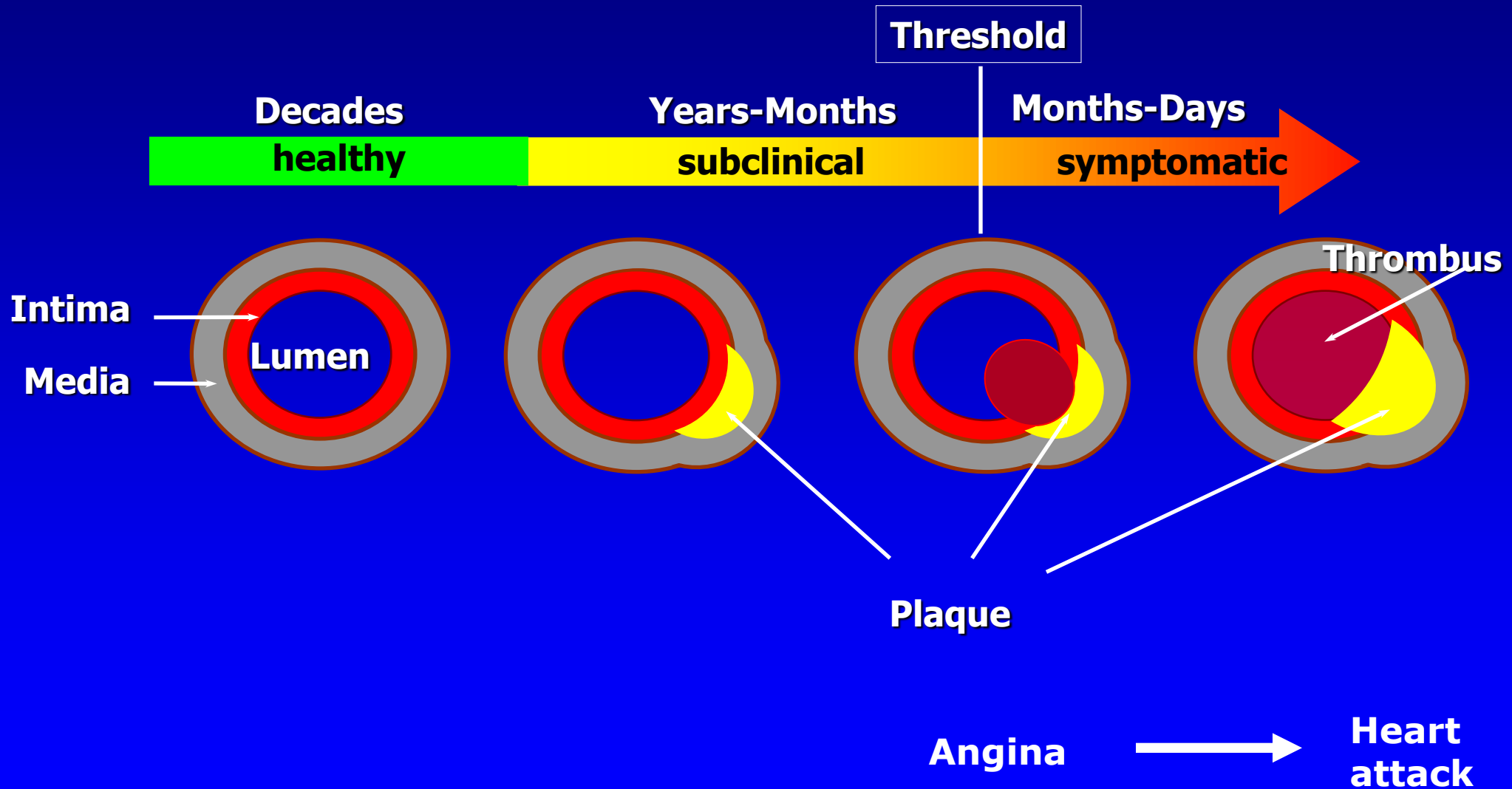
The questions

- Why troponin is such a good biochemical marker (what happens when there is cellular destruction)
- The typical pattern of troponin increase following MI
- Why it is unlikely troponin will be replaced in the near future by any other biochemical marker
- Why the new hs troponins are much better – and why more accurate levels can be taken sooner than with the older assays
- How long have the hs troponins been used for in the UK and how long will it be before they are used in all of the UK hospitals

What we used to think



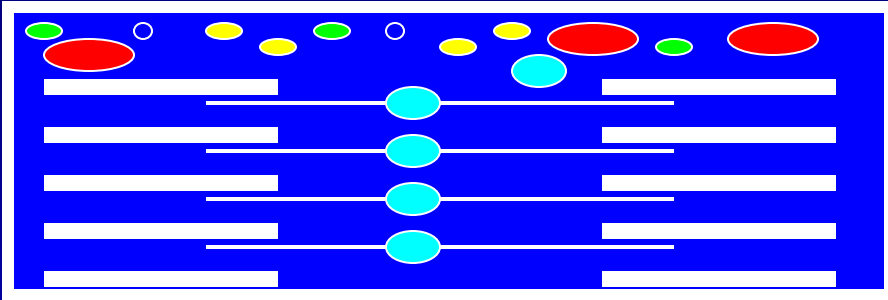
What we now think



The questions

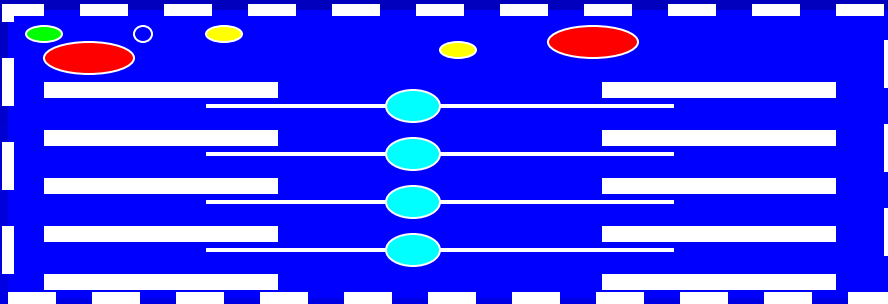
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- The typical pattern of troponin increase following MI

Normal



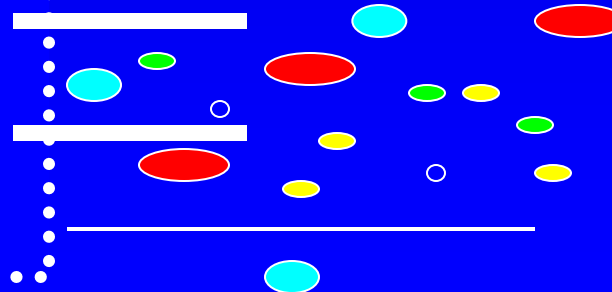
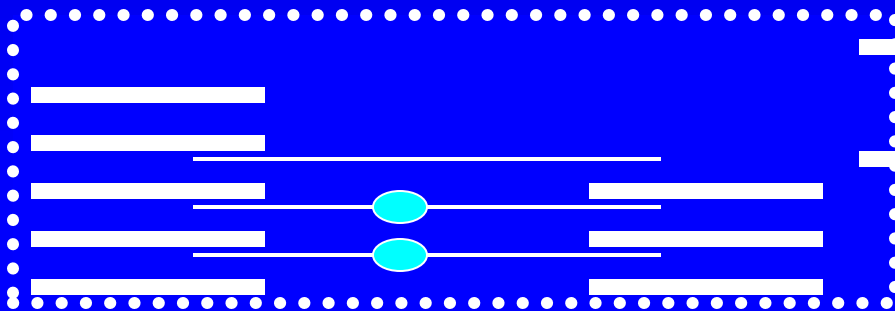
Background levels represent cellular leakage/degradation

Ischaemia/Early infarction



Leakage due to loss of membrane integrity
?upregulation of soluble cytosolic forms or
intracellular degradation and release of small
peptides

Infarction



Release of cytosolic
and structural
proteins

The ideal cardiac marker

- Completely cardiac specific
- Not found in the normal circulation
- Stable in storage
- Well characterised
- Clinical role defined
- Easy to measure
- Tax deductible
- Tastes of chocolate

Markers of myocardial damage

- Cytosolic

- AST
- LDH
- LD isoenzymes (LD1, LD1/2 ratio, HBD)
- CK
- CK MM and MB isoforms
- CK-MB (activity)
- CK-MB (concentration)
- Myoglobin
- Glycogen phosphorylase BB (GPBB)
- Heart fatty acid binding protein (HFABP)

- Structural proteins

- Troponin T
- Troponin I

Troponin: Biochemistry

The troponin complex is a component of the thin filaments

in striated muscle complexed to actin

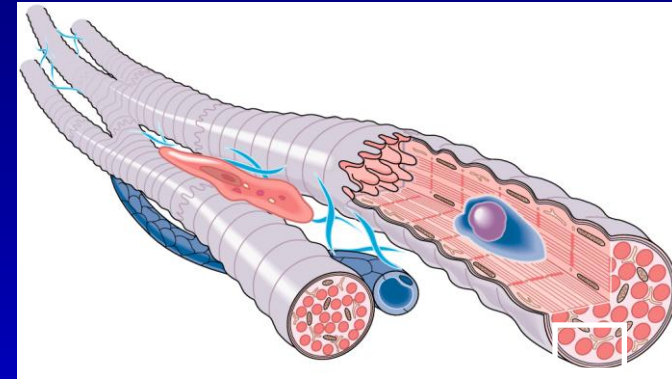
There are three types of troponins:

- Troponin T (Tropomyosin binding)
- Troponin I (inhibitory protein)
- Troponin C (Calcium Binding)

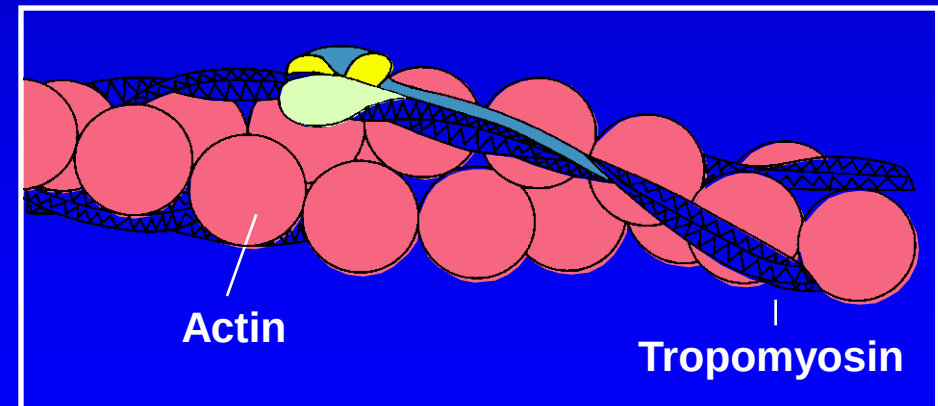


The troponins are three different proteins structurally not related with each other

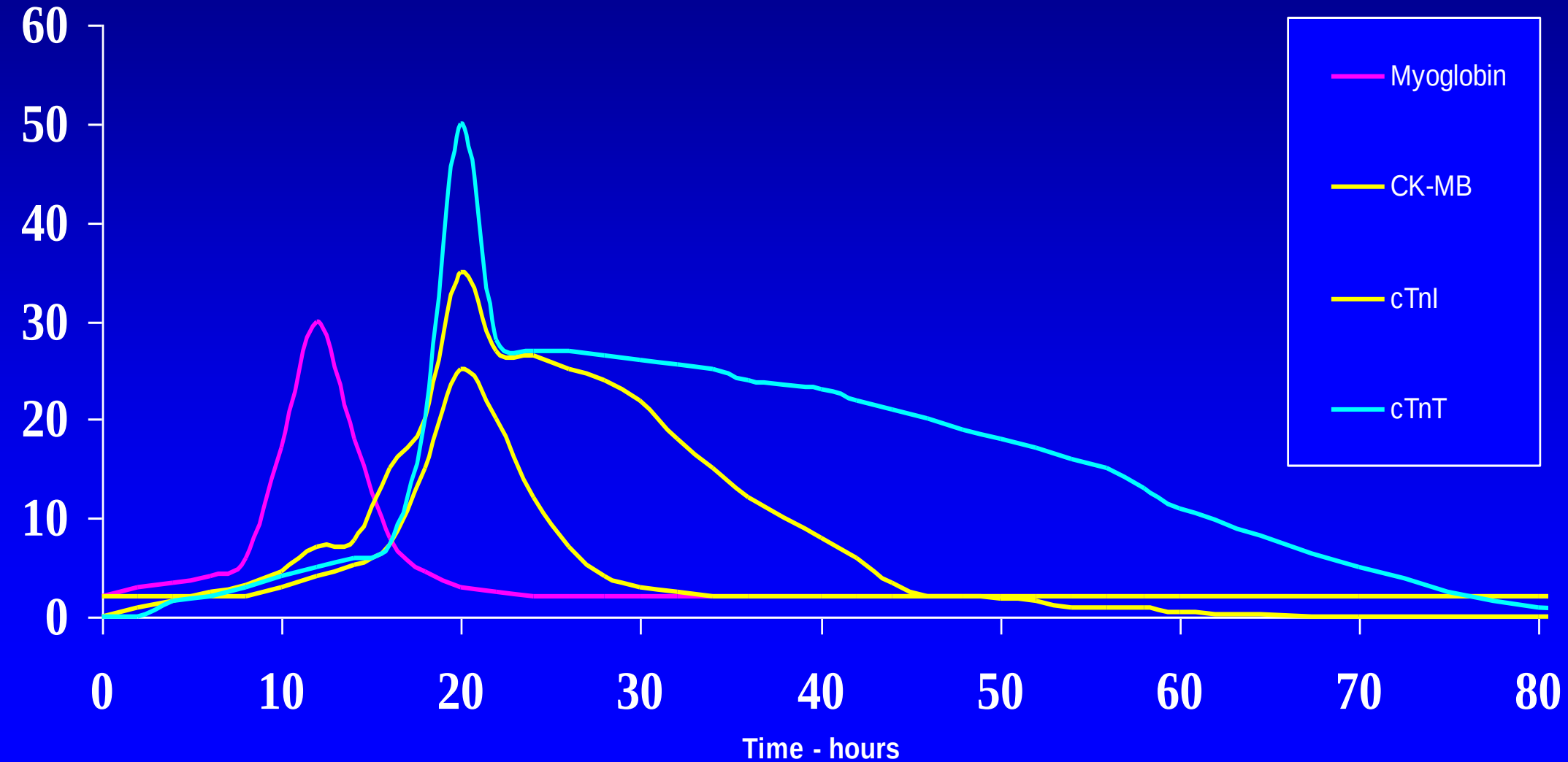
Cardiac troponin T and I differ significantly from troponin T and I found in skeletal muscle



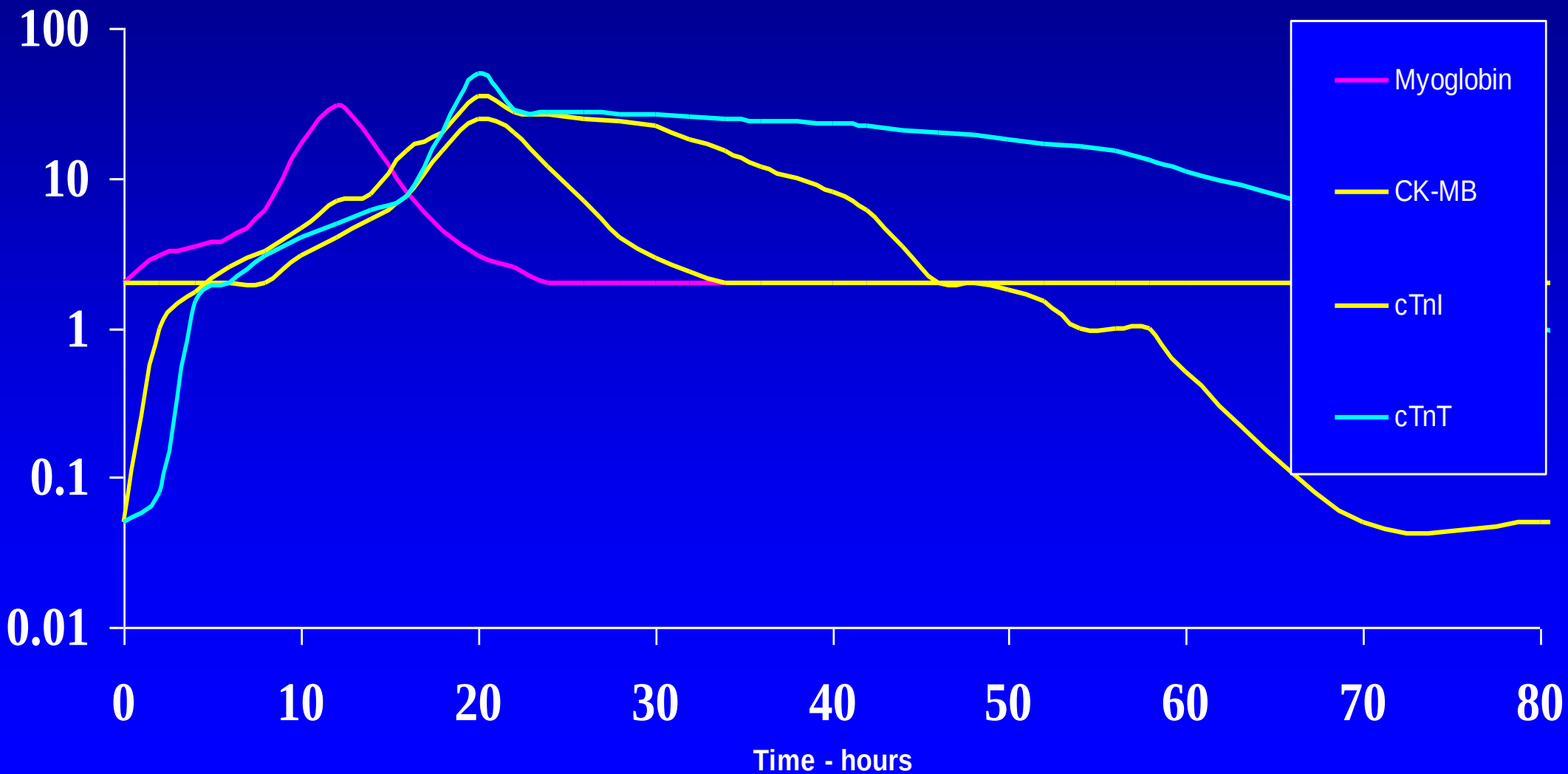
Cardiac myofiber



Release kinetics



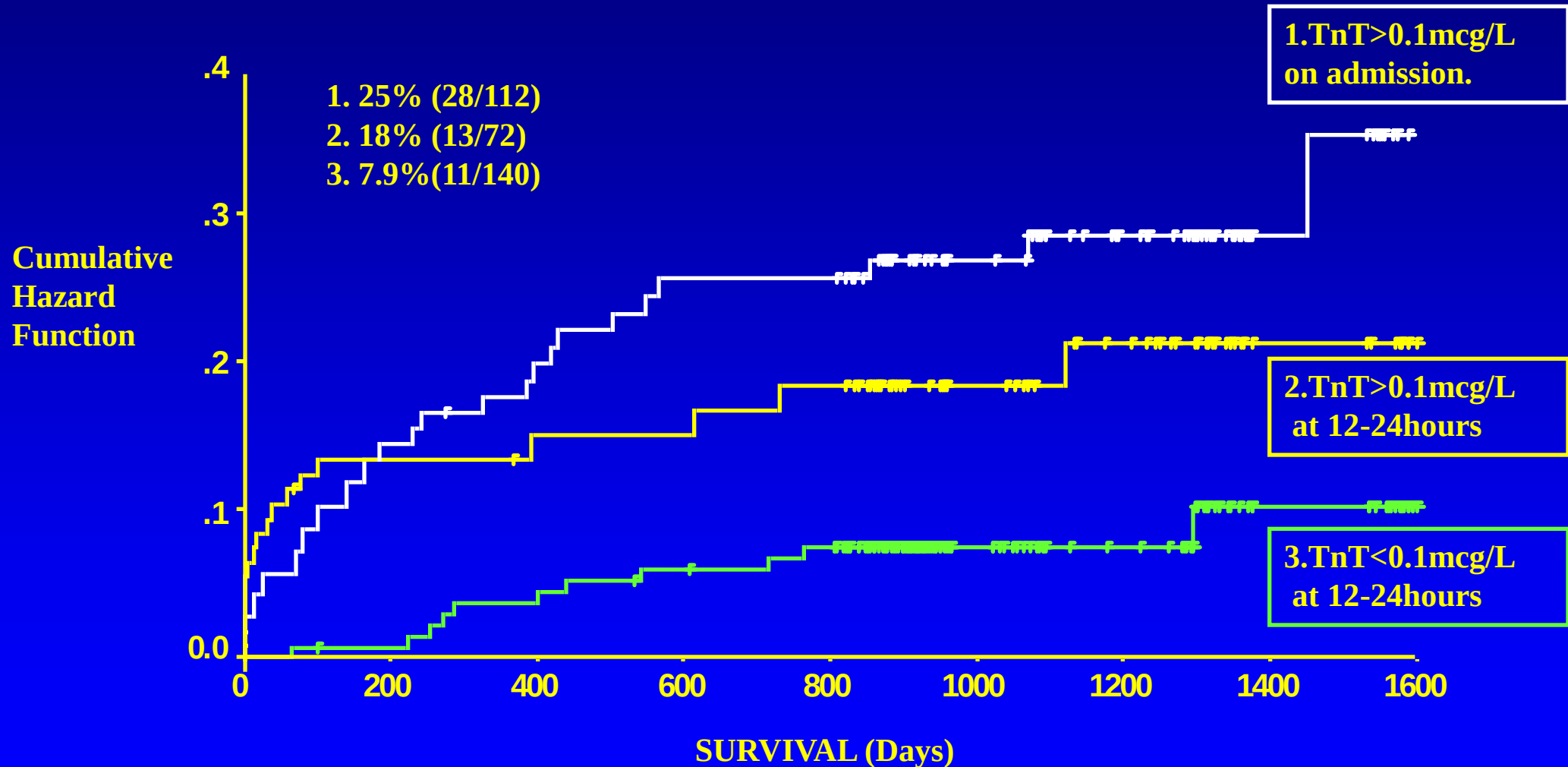
Release kinetics



Advantages of cTn over other markers

- Cardiospecific
- Sensitive - diagnosis using CK-MB is flawed. Elevation of cTn in patients “without AMI”
 - occurs in 33% of patients
 - indicates myocardial damage
 - predicts major adverse cardiac events (MACE)
 - death
 - MI
 - readmission with UAP
 - need for urgent revascularisation

Cardiac death as first event according to timing of Troponin T measurement and concentration in patients admitted with chest pain without ST segment elevation.



Log rank: 1 v2 p=0.27, 1 v3 p=0.0002, 2 v3 p=0.026

Why troponin

- Troponin is one of two truly evidence based laboratory tests.
- It is both diagnostic and prognostic
- This has been confirmed by meta-analysis

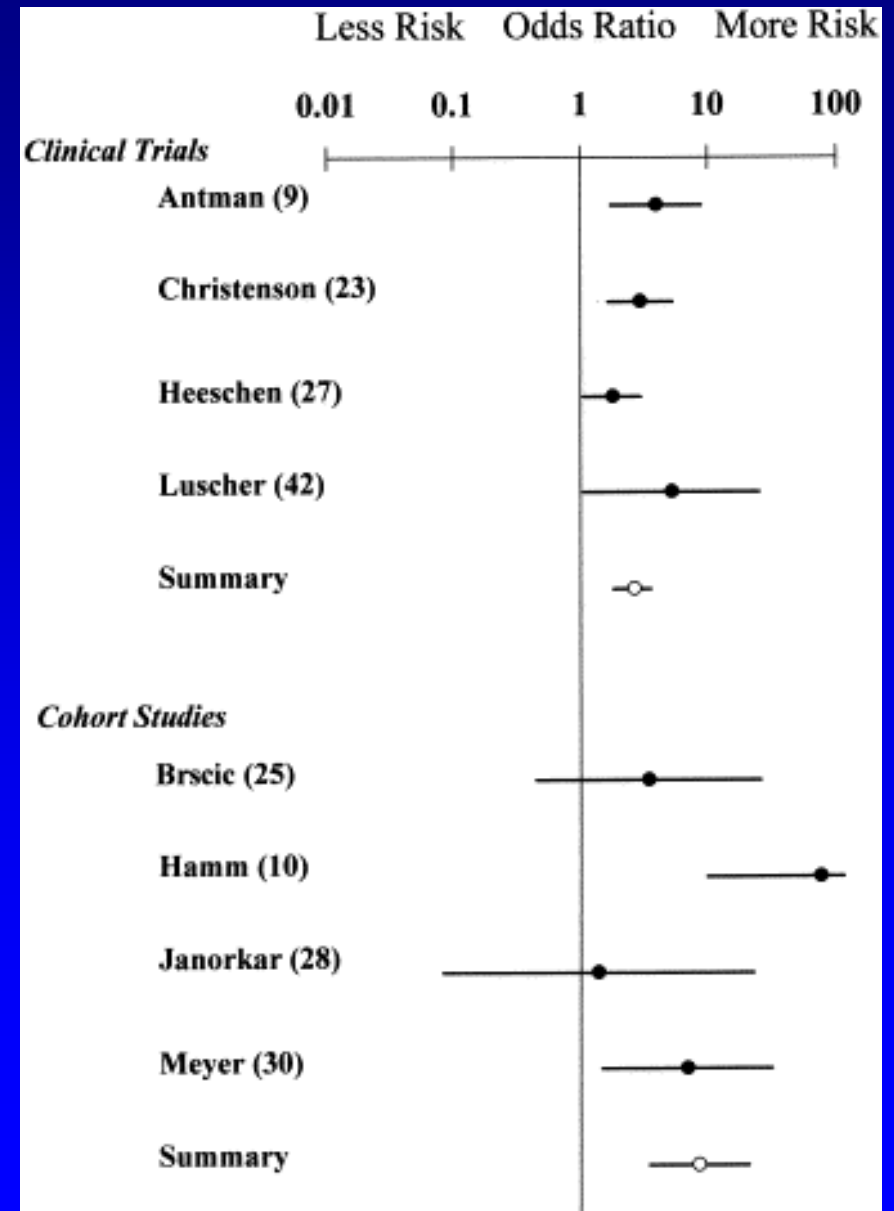
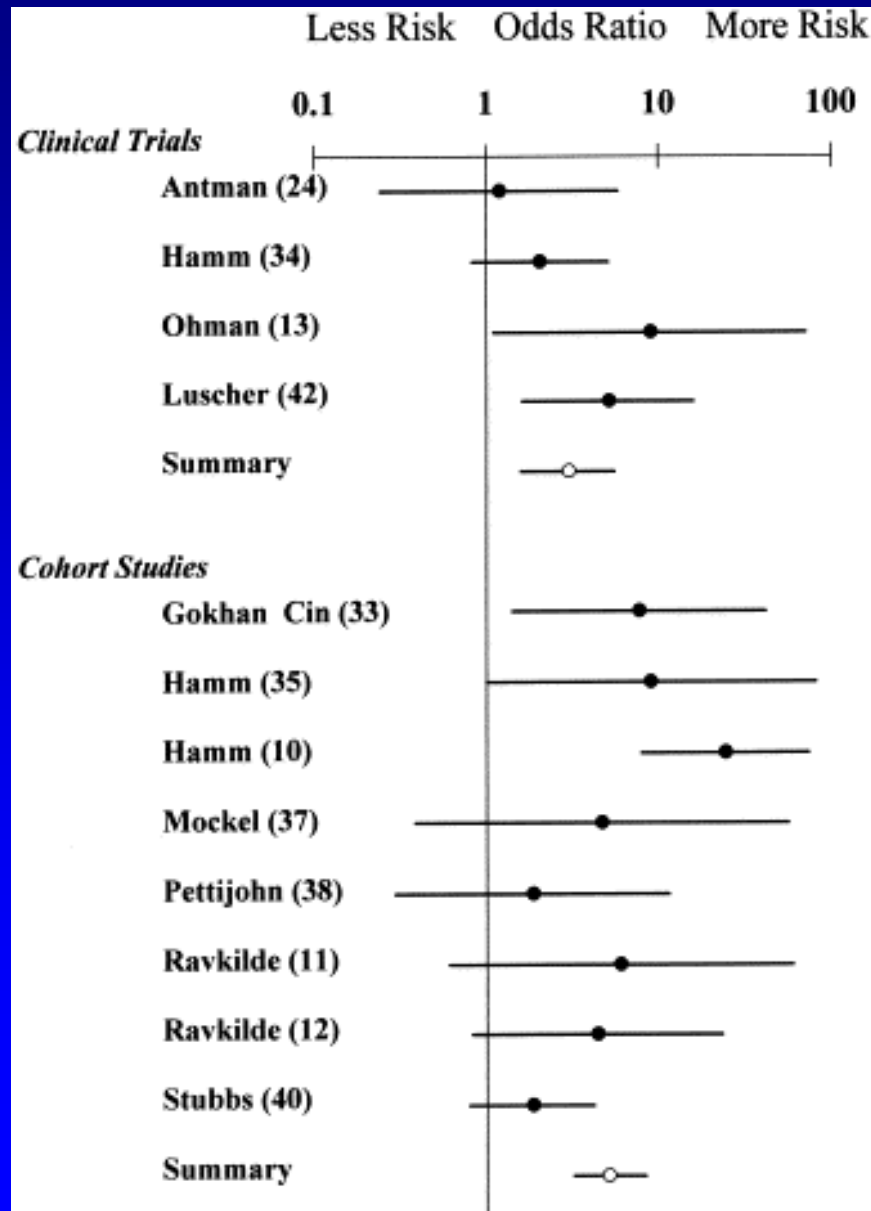
Olatidoye AG, Wu AH, Feng YJ, Waters D: Prognostic role of troponin T versus troponin I in unstable angina pectoris for cardiac events with meta-analysis comparing published studies. *Am.J.Cardiol.* 1998, **81**:1405-1410.

Heidenreich PA, Alloggiamento T, Melsop K, McDonald KM, Go AS, Hlatky MA. The prognostic value of troponin in patients with non-ST elevation acute coronary syndromes: a meta-analysis. *J.Am.Coll.Cardiol.* 2001;**38**:478-85.

Fleming SM, Daly KM. Cardiac troponins in suspected acute coronary syndrome: a meta-analysis of published trials. *Cardiology* 2001;**95**:66-73.

Balk EM, Ioannidis JP, Salem D, Chew PW, Lau J. Accuracy of biomarkers to diagnose acute cardiac ischemia in the emergency department: a meta-analysis. *Ann.Emerg.Med.* 2001;**37**:478-94.

Meta-analysis data for cTnT (left) and cTnI (right) adapted from Heidenreich PA et al *J.Am.Coll.Cardiol.* 2001;38:478-85



The questions

- Why it is unlikely troponin will be replaced in the near future by any other biochemical marker

Diagnosis of myocardial infarction

What has troponin ever done for us?



Consensus Document

Myocardial infarction redefined — A consensus document of The Joint European Society of Cardiology/American College of Cardiology Committee for the Redefinition of Myocardial Infarction

**The Joint European Society of Cardiology/American College of Cardiology
Committee****

Universal definition of myocardial infarction

Kristian Thygesen, Joseph S. Alpert and Harvey D. White on behalf of the Joint ESC/ACCF/AHA/WHF Task Force for the Redefinition of Myocardial Infarction

Third universal definition of myocardial infarction

Kristian Thygesen, Joseph S. Alpert, Allan S. Jaffe, Maarten L. Simoons, Bernard R. Chaitman and Harvey D. White: the Writing Group on behalf of the Joint ESC/ACCF/AHA/WHF Task Force for the Universal Definition of Myocardial Infarction

Definition of myocardial infarction

Criteria for acute myocardial infarction

The term acute myocardial infarction (MI) should be used when there is evidence of myocardial necrosis in a clinical setting consistent with acute myocardial ischaemia. Under these conditions any one of the following criteria meets the diagnosis for MI:

- Detection of a rise and/or fall of cardiac biomarker values [preferably cardiac troponin (cTn)] with at least one value above the 99th percentile upper reference limit (URL) and with at least one of the following:
 - ♦ Symptoms of ischaemia.
 - ♦ New or presumed new significant ST-segment–T wave (ST–T) changes or new left bundle branch block (LBBB).
 - ♦ Development of pathological Q waves in the ECG.
 - ♦ Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality.
 - ♦ Identification of an intracoronary thrombus by angiography or autopsy.
- Cardiac death with symptoms suggestive of myocardial ischaemia and presumed new ischaemic ECG changes or new LBBB, but death occurred before cardiac biomarkers were obtained, or before cardiac biomarker values would be increased.

Classification of myocardial infarction

- **Type 1** – classical AMI – Spontaneous myocardial related to ischaemia due to a primary coronary event such as plaque erosion or rupture, fissuring or dissection.
- **Type 2** – Myocardial infarctions secondary to ischaemia due to imbalance between oxygen demand and supply e.g. coronary spasm, anaemia or hypotension.
- **Type 3** – Sudden cardiac death with symptoms of ischaemia, accompanied by new ST elevation or LBBB, or verified coronary thrombus by angiography or autopsy, and death occurring before blood samples could be obtained.
- **Type 4a** - Myocardial infarction associated with PCI.
- **Type 4b** - Myocardial infarctions associated with verified stent thrombosis.
 - Defined as biomarkers exceeding five times the upper reference limit.
- **Type 5** -Myocardial infarction associated with CABG.
 - Defined as biomarkers exceeding ten times the upper reference limit.

WHO to the new definition – evolution or revolution?

Q wave AMI - Definite
ECG and cardiac enzyme
changes

Non Q wave AMI - ECG
changes not definitive but
definite cardiac enzyme changes

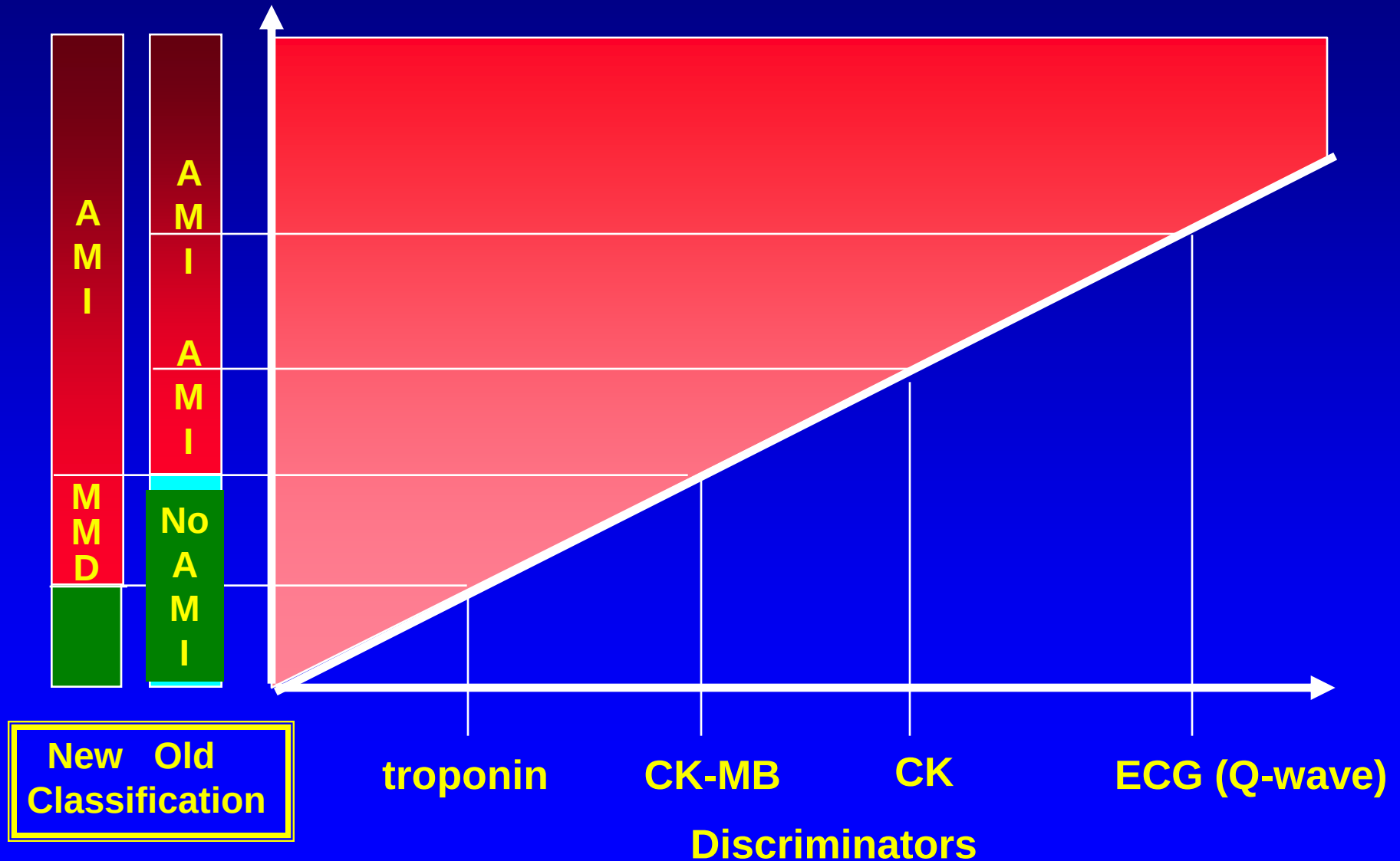
Unstable angina – ECG and
cardiac enzyme changes not
definitive

ST elevation MI (STEMI)
- Definite ECG and cardiac
troponin changes

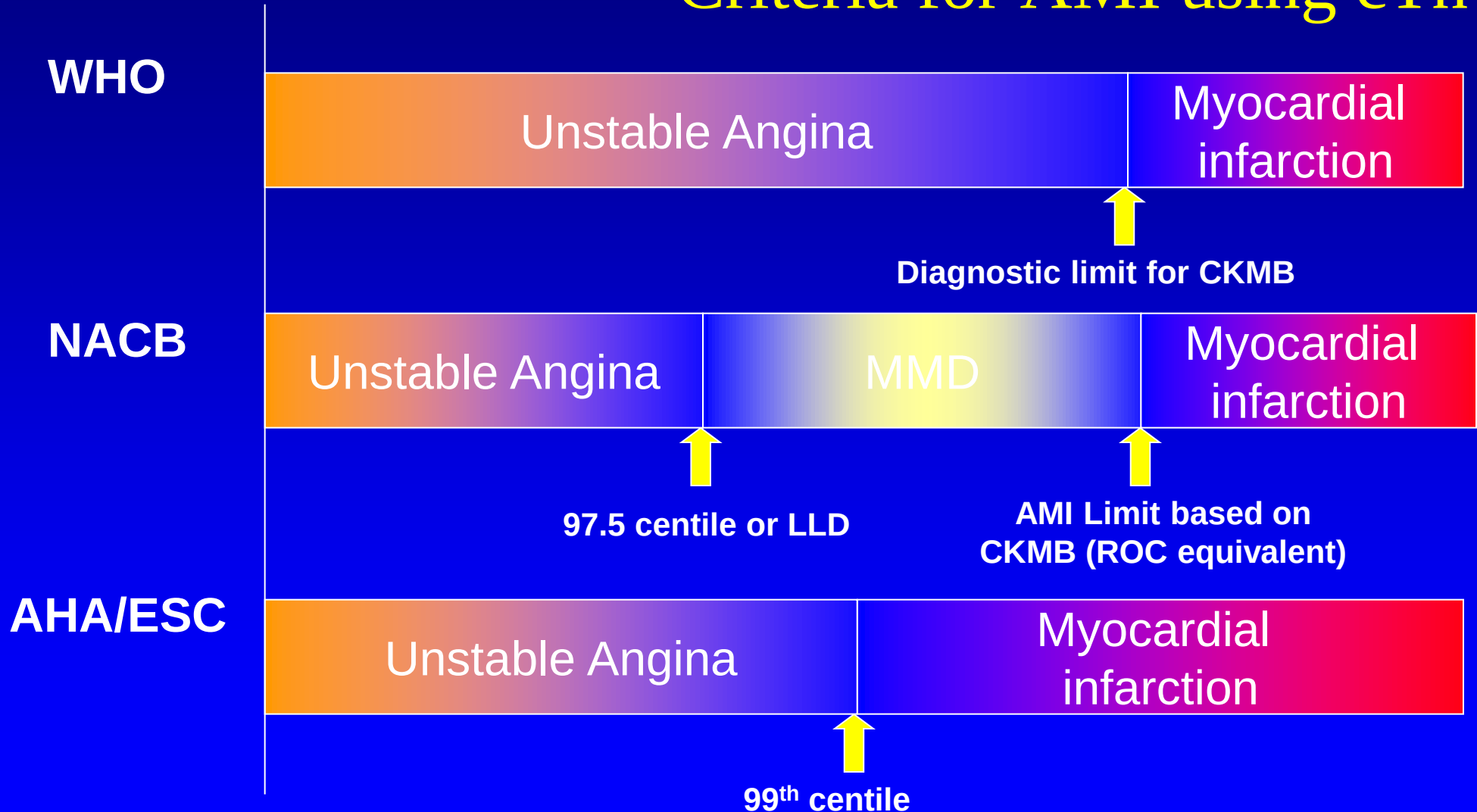
Non ST elevation MI
(NSTEMI) - ECG changes not
definitive but definite cardiac
troponin changes

Unstable angina – ECG
changes not definitive and no
detectable cardiac troponin

WHO to the new definition – evolution or revolution?



Evolution of Diagnostic Criteria for AMI using cTn



New vs Old

- For all the existing cardiac troponin methods a cut off is defined that is equivalent to the previous WHO AMI cut offs

Method	99 th centile	AMI cut off	
		Old units	New units
Cardiac troponin I			
Abbott AxSym	0.4 (95 th centile)	2.0	2000
Abbott ARCHITECT	0.012	0.3	300
ADVIA Centaur cTnI	0.07 (95 th centile)	1.5	1500
ADVIA Centaur TnI-Ultra	0.04	0.78	780
Access Accu TnI	0.04	0.5	500
Dade Behring Dimension	0.07	0.6	600
Dade Behring Stratus	0.07	0.6	600
IMMULITE 2500	0.2	0.9	900
Ortho VITROS cTnI	0.08	0.4	400
Ortho VITROS Tropl		0.12	120
Tosoh AIA	0.1	0.1	100
Cardiac troponin T			
Roche	0.14	0.1	100

The questions

- Why the new hs troponins are much better – and why more accurate levels can be taken sooner than with the older assays

What are “sensitive” troponin assays

Conventional troponin

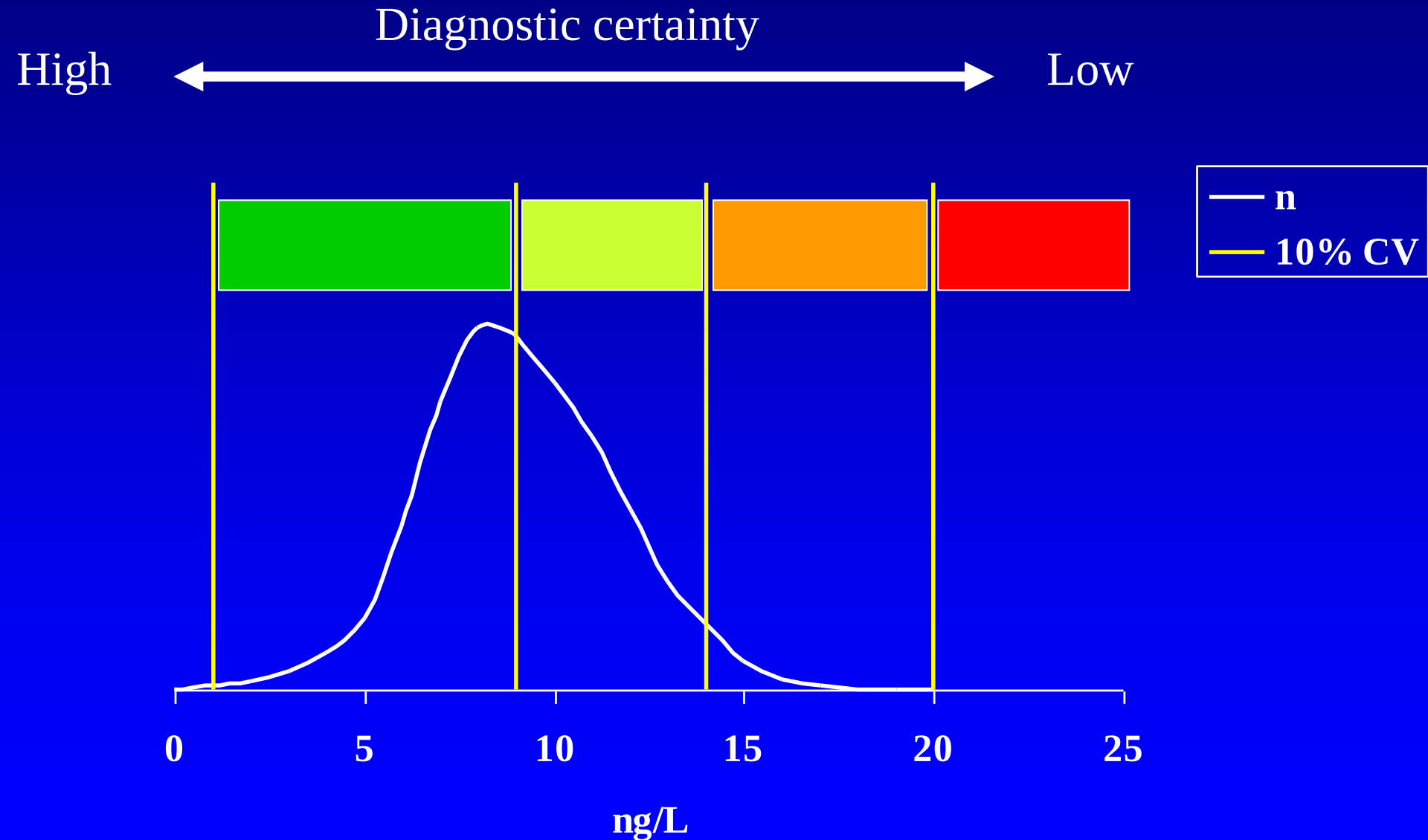


Or new troponin



And how expensive is the change – a new high maintenance assay?

Assay performance to achieve a true 99th centile



Scorecard designations of cTn assays.

Acceptance designation	Total imprecision at the 99th percentile, CV%
Guideline acceptable	<10
Clinically usable	≥ 10 to ≤ 20
Not acceptable	>20
Assay designation	Measurable normal values below the 99th percentile, %
Level 4 (third generation, hs)	≥ 95
Level 3 (second generation, hs)	75 to < 95
Level 2 (first generation, hs)	50 to < 75
Level 1 (contemporary)	<50

Adapted from Apple Clin Chem 2009

Company/platform/assay ¹	99th percentile, µg/L	10% CV, µg/L	Acceptance designation	Assay designation
Abbott Architect	0.028	0.032	Clinically usable	Level 1
Beckman Access Accu	0.04	0.06	Clinically usable	Level 2
bioMerieux Vidas Ultra	0.01	0.11	Not acceptable	Level 1
Ortho-Clinical Diagnostics Vitros ECi ES	0.034	0.034	Guideline acceptable	Level 1
Radiometer AQT90	0.023	0.039	Clinically usable	Level 1
Roche Elecsys 2010	<0.01	0.03	Clinically usable	Level 1
Siemens Centaur Ultra	0.04	0.03	Guideline acceptable	Level 1
Siemens Dimension RxL	0.07	0.14	Clinically usable	Level 1
Siemens Immulite 2500 STAT	0.2	0.42	Not acceptable	Level 1
Siemens Stratus CS	0.07	0.06	Guideline acceptable	Level 1
Siemens VISTA	0.045	0.04	Guideline acceptable	Level 1
Tosoh AIA II	<0.06	0.09	Clinically usable	Level 1
Research hs assays				
Beckman Access hs-cTnI	0.0086	0.0086	Guideline acceptable	Level 4
Roche Elecsys hs-cTnT	0.013	0.012	Guideline acceptable	Level 4
Nanosphere hs-cTnI	0.0028	0.0005	Guideline acceptable	Level 3
Singulex hs-cTnI	0.0101	0.00088	Guideline acceptable	Level 4

Adapted from Apple Clin Chem 2009

What is a high sensitivity assay

- A high sensitivity assay refers to the assays analytical characteristics, it does not mean a different form of troponin is being measured
- Total imprecision at the 99th percentile should be $\leq 10\%$
- The assay should detect at least 50% of healthy individuals above the LoD of the assay
- Results should be reported in ng/L

Time to diagnosis

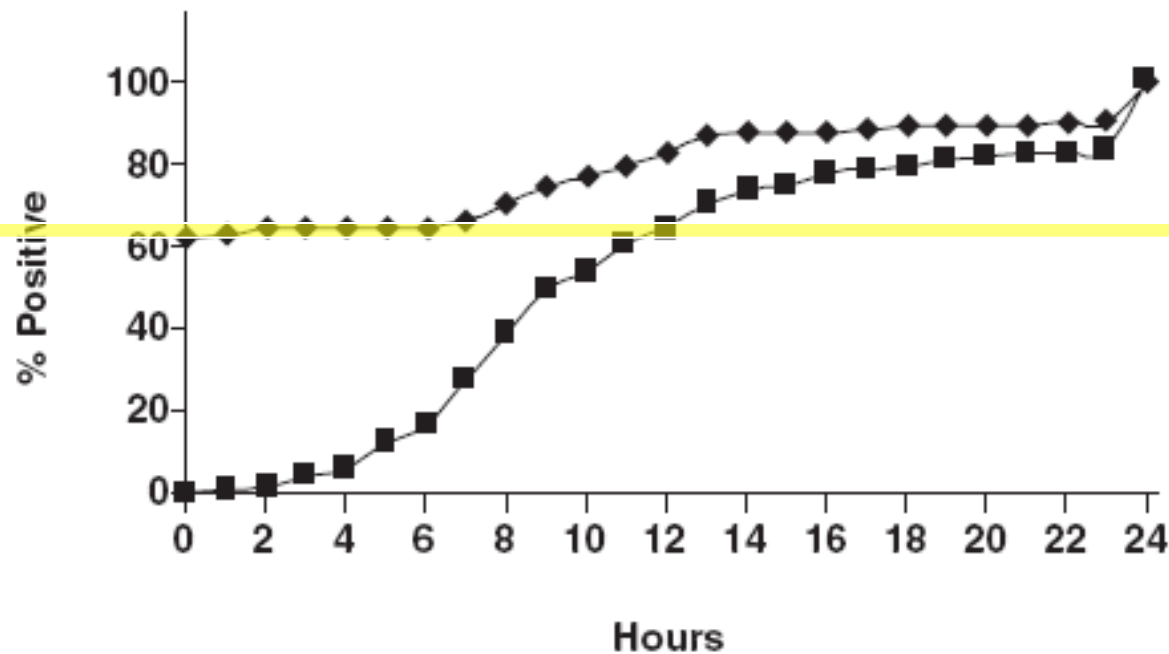
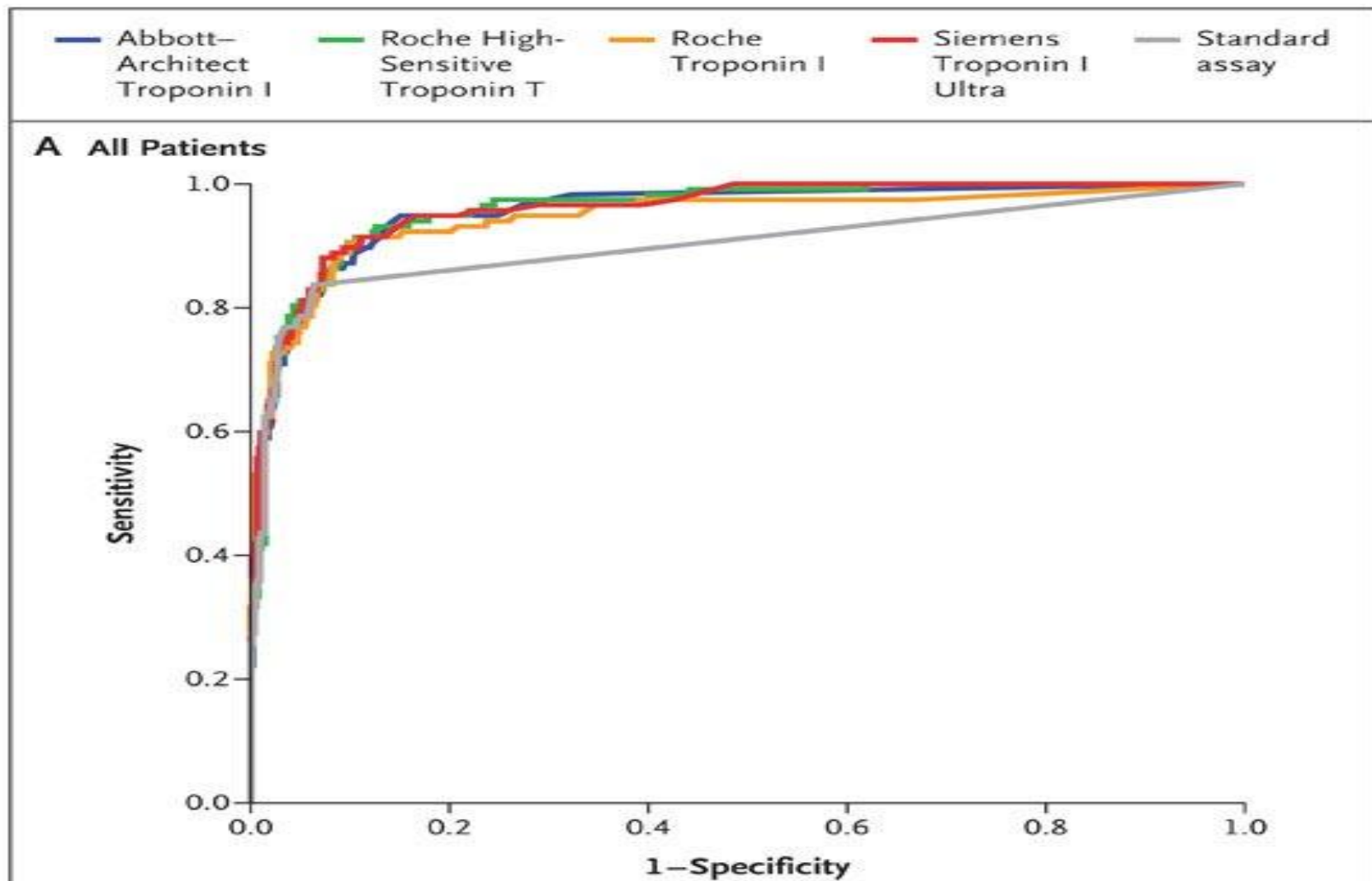
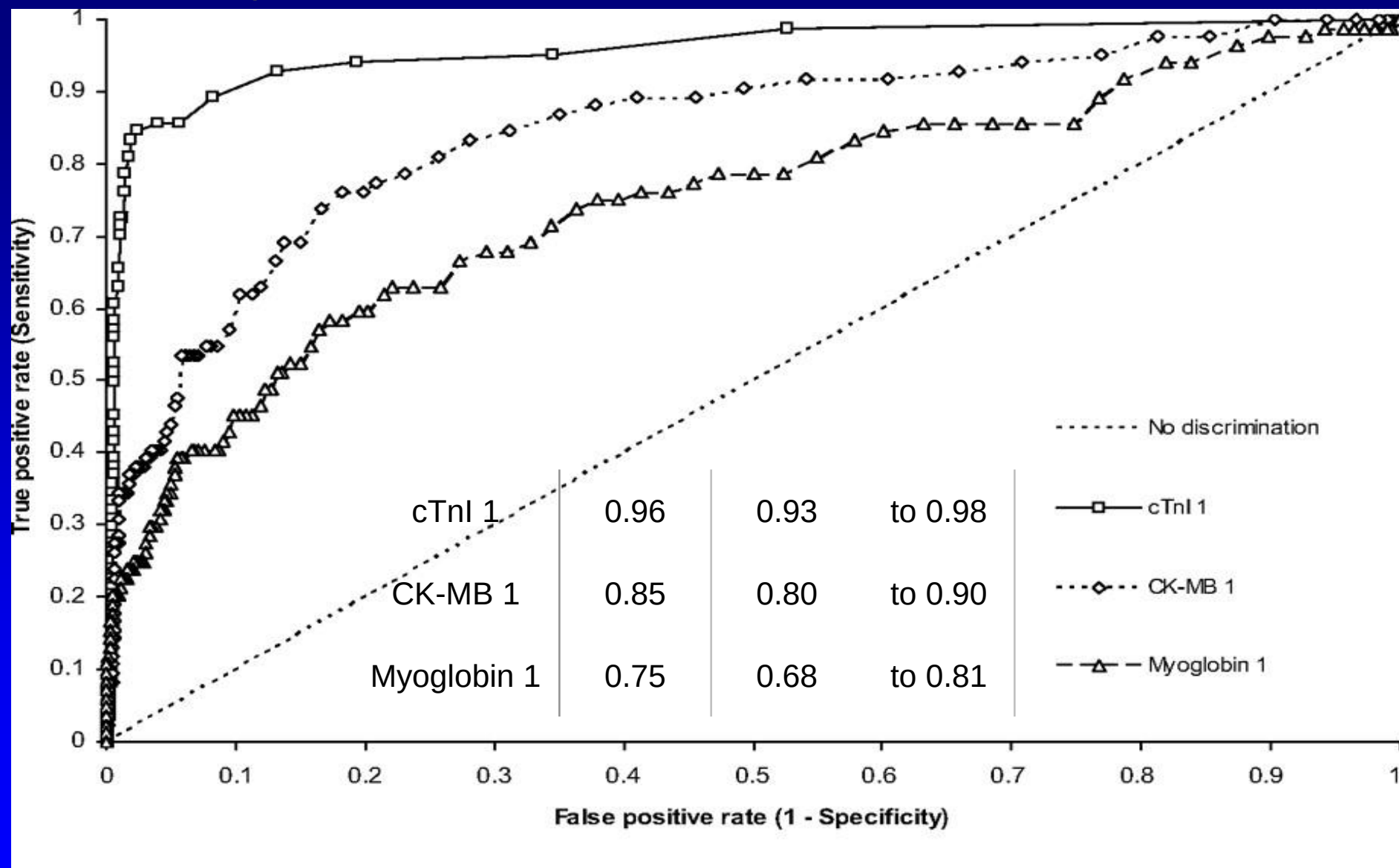


Figure 2 Timing of conversion to positive troponin I (TnI) in patients with myocardial injury. The percentage of patients with positive TnI-Ultra (diamonds) and positive cardiac TnI (cTnI; squares) is shown starting when the baseline specimen was obtained (ie, 0 hours) in a 24-hour period. By study design, all initial specimens for cTnI are negative.

Diagnostic Performance of Cardiac Troponin Assays at Presentation



Receiver operator characteristic curves for cardiac biomarkers for the diagnosis of acute myocardial infarction on admission.



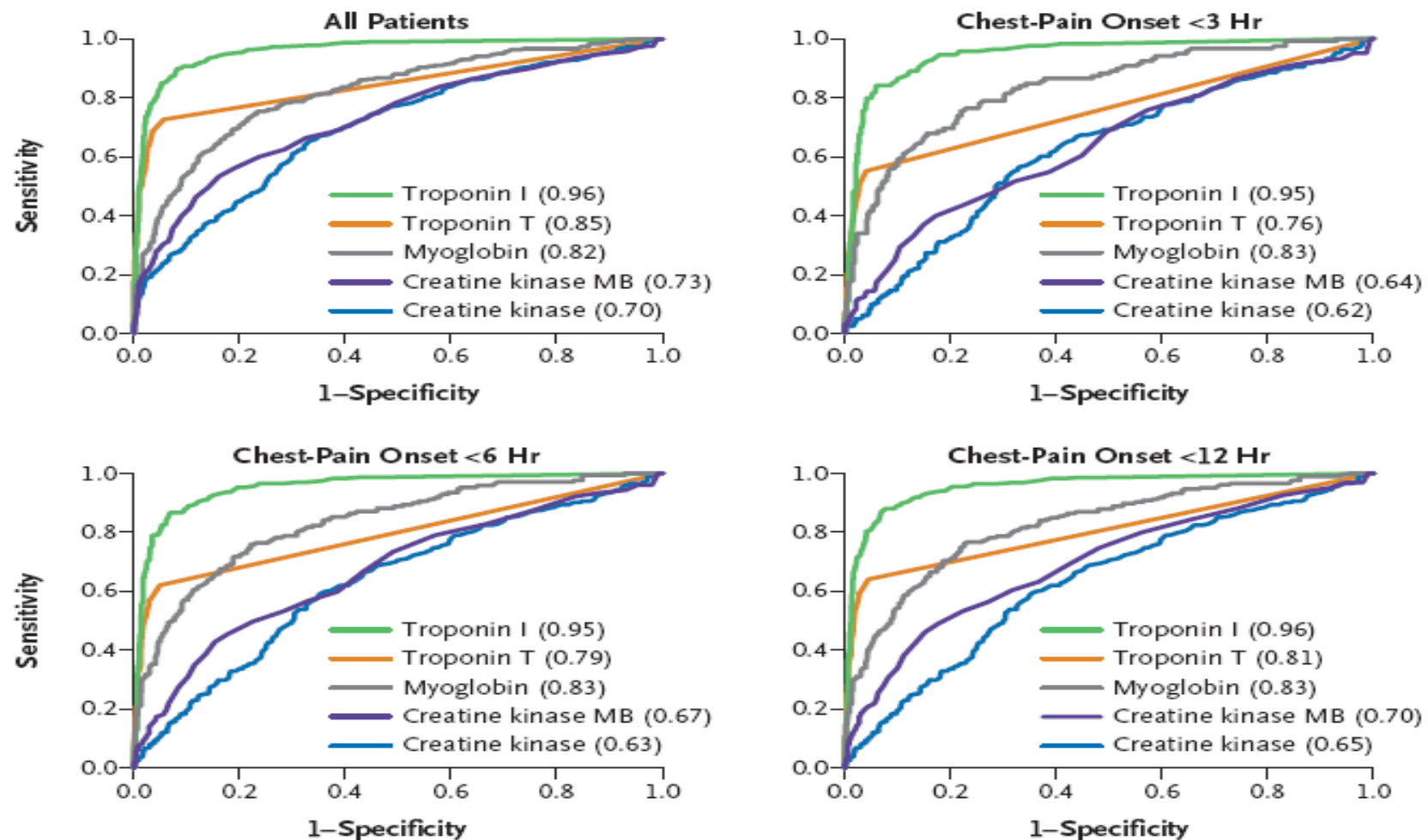


Figure 1. Diagnostic Accuracy of Single Biomarker Testing for Acute Myocardial Infarction.

Shown are receiver-operating-characteristic curves and the corresponding areas under the curve for baseline measurements of troponin I with the sensitive assay (troponin I), troponin T, myoglobin, creatine kinase MB, and creatine kinase, according to the time of the onset of chest pain. Also shown are the sensitivity and specificity of these measures.

Troponin elevations outside ACS

Troponin is the answer to all diagnostic problems
Its not the messiah, its just a very naughty molecule



Classification of myocardial infarction

- **Type 1** – classical AMI – Spontaneous myocardial related to ischaemia due to a primary coronary event such as plaque erosion or rupture, fissuring or dissection.
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Myocardial Infarction Redefined

A clinical framework for interpretation

CLINICAL : TROPONIN T and TROPONIN I
Troponin Positive in unselected patients

PICI: Primary Ischaemic Cardiac Injury due to an ACS
(Primarily due to a Ruptured Coronary Plaque)

SICI: Secondary Ischaemic Cardiac Injury
(NOT Primarily due to a Ruptured Coronary Plaque)

NICI: Non Ischaemic Cardiac Injury

Classification of myocardial infarction - PICI

- **Type 1** – classical AMI – Spontaneous myocardial related to ischaemia due to a primary coronary event such as plaque erosion or rupture, fissuring or dissection.
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Classification of myocardial infarction

- **Type 2** – Myocardial infarctions secondary to ischaemia due to imbalance between oxygen demand and supply e.g. coronary spasm, anaemia or hypotension.

SICI: Secondary Ischaemic Cardiac Injury (type 2 AMI)

- Sympathomimetics
 - Cocaine (Chicago)
 - ‘Catecholamine Storm’-
 - Head Injury, Stroke, intracerebral haemorrhage
- Pulmonary Embolus*
 - Presumed Right Heart Strain or Hypoxia
- Coronary Artery Spasm
- Coronary Artery Embolisation
 - Clot, Air, CABG
- Coronary Artery Inflammation with microvascular occlusion
 - Connective tissue disease
 - Vasculitidies
 - Kawasaki disease
 - Sarcoid
- Hypotension

SICI: Secondary Ischaemic Cardiac Injury (Type 2 AMI)

- End Stage Renal Failure
 - More Severe CAD (50% may normal coronaries)
- Rhythm Disturbances
 - Prolonged Tachy or Brady(? Only if have IHD)
- Acute Heart Failure (?Only if IHD aetiology)
- Chronic Heart Failure
- Aortic Stenosis and HOCM
- Direct Coronary Artery Trauma
- Extreme Marathons
 - Wall motion abnormalities
- Extreme endurance
 - cTn +ve deaths presumed due to extreme Oxygen debt and Ischaemia

SICI: Secondary Ischaemic Cardiac Injury

Troponin +ve not due to primary plaque rupture (type 4-5 MI)

- **Coronary Interventions**

- **PCI**

 - **Primary**

 - **Clot/ Atheroma distal embolisation**

 - **Side Branch occlusion**

- **Elective**

 - **Distal embolisation from atheroma/debris**

 - **Side Branch occlusion**

- **CABG**

 - **Global Ischaemia from inadequate perfusion, myocardial cell protection or anoxia**

NICI: Non Ischaemic Cardiac Injury

- **Known Causes of Myocarditis**

- **Infection- Bacterial, Viral (Coxsackie, parvovirus B 19)**

- **Cardiac Trauma**

- **Surgery, contusion, ablation, pacing, defibrillators, cardioversion, endomyocardial biopsy, closure of ASD's**

- **Inflammation**

- **Kawasaki disease, myocardial extension of bacterial endocarditis**

- **Auto-Immune**

- **Drugs**

- **Alcohol, Chemotherapy (adriamycin, 5 fluorouracil), Carbon monoxide, snakebite, pufferfish poisoning.**

-

NICI: Non Ischaemic Cardiac Injury

- Rhabdomyolysis with cardiac injury
- Infiltrative diseases
 - Polymyositis, scleroderma, sarcoidosis, haemochromatosis, glycogen storage disease.
- Renal Failure
- Critically ill patients
 - especially with diabetes, respiratory failure, sepsis, burns, multiorgan failure.

Mechanisms in SICI and NICI

- From direct myocardial injury to vascular injury plus ischaemia
 - Direct myotoxicity
 - drugs, envenomation, myocarditis
 - Vascular injury
 - Connective tissue disease
 - Supply demand mismatch
 - Vasospasm (cocaine)
 - Low flow watershed infarction

The questions

- How long have the hs troponins been used for in the UK and how long will it be before they are used in all of the UK hospitals
 - Siemens ultra assay for the past 3 years (new version soon)
 - Roche assay now
 - Beckmann now
 - Abbott due for launch
- They have become the standard of care

